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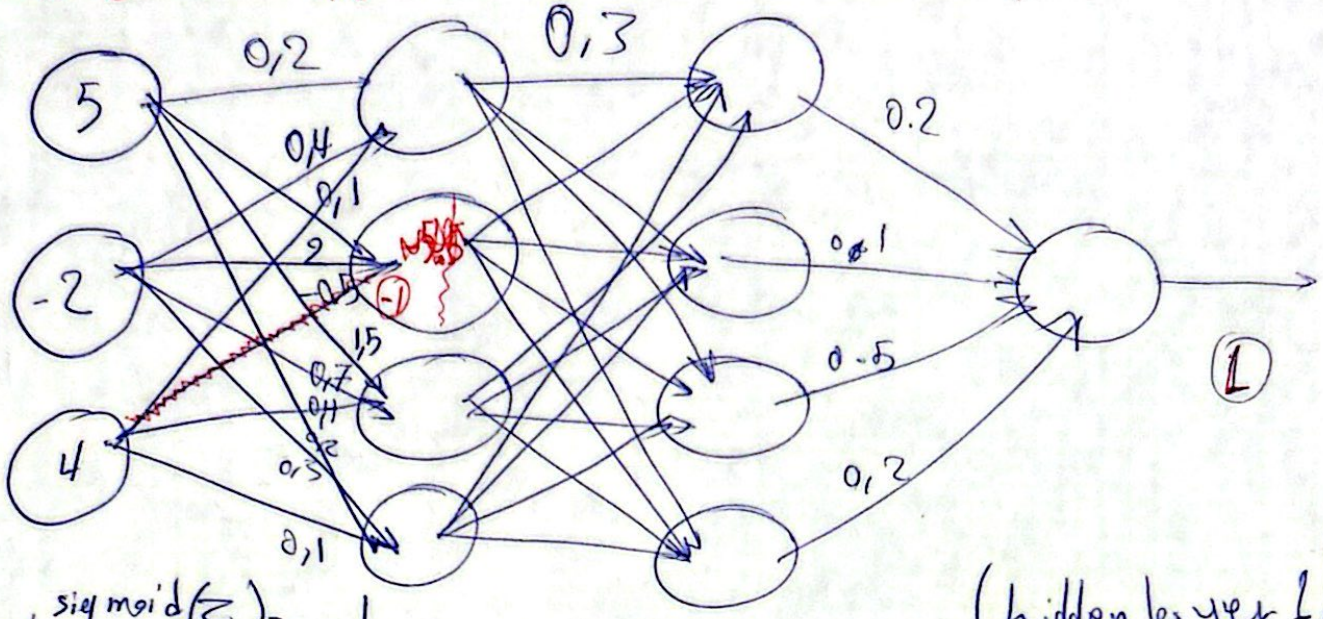


(hidden layer 1)

forward $\text{sigmoid}(z) = \frac{1}{1+e^{-z}}$

Node 2:

$$z_2^{(1)} = (5 \times 1) + (-2 \times 2) + (4 \times -0.5) = 5 - 4 - 2 = \cancel{5.5}\ (-1)$$

$$a_2^{(1)} = \frac{1}{1+e^{-(-1)}} = \boxed{0.2689}$$

Node 1: $(5 \times 0,2) + (-2 \times 0,4) + (4 \times 0,1) = 1 - 0,8 + 0,4 = 0,6 \Rightarrow a_1^{(1)} = \frac{1}{1+e^{-0,6}} = \boxed{0,6457}$

Node 3: $z_3^{(1)} = (5 \times 0,7) + (-2 \cdot 1,5) + (4 \cdot 0,4) = 3.5 - 3.0 + 1,6 = 2.1 \Rightarrow a_3^{(1)} = \frac{1}{1+e^{-2,1}} = \boxed{0,8909}$

Node 4: $z_4^{(1)} = (5 \times 0,3) + (-2 \cdot 0,3) + (4 \cdot 0,3) = 1,5 - 0,4 + 0,4 = 1.5 \Rightarrow a_4^{(1)} = \frac{1}{1+e^{-1,5}} = \boxed{0,8176}$

(h-layer 2)

$$z^{(2)} = 0,3 \times (0,6457) + 0.3 \times (0,8909) + 0,3 (0.8176)$$

$$z^{(2)} = 0,3 (0,6457 + 0,2686 + 0.8909 + 0,8176) = 0,3 (2.6231) = 0,7869$$

$$a^{(2)} = \frac{1}{1+e^{-0,7869}} = \boxed{0,6872}$$

output layer: $z^{(3)} = (0.2 \times 0,6872) + (0,1 \times 0,6872) + (0,5 \times 0,6872) + 0,2 \times 0,6872)$

$$z^{(3)} = (0,2 + 0,1 + 0,5 + 0,2) \cdot 0,6872 = 1.0 \times 0,6872 = 0,6872 \quad a = \frac{1}{1+e^{-0,6872}} = \boxed{0,6654}$$

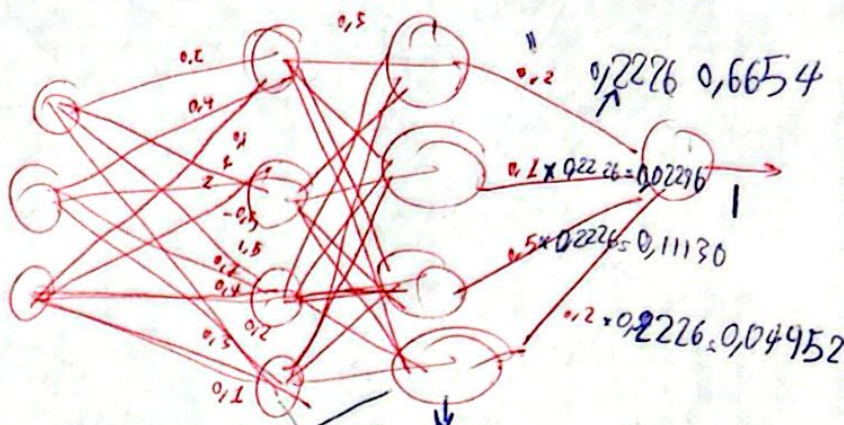
Scanned with CamScanner

Backpropagation:

sigmoid gradient $= \frac{d\sigma(x)}{dx} = (1-\sigma(x))\,\sigma(x)$

$0.6654\,(1-0,6654) = 0,2226$

$0,2226 \times 0,2 = 0,04452$



local gradient sigmoid

$0,6872(1-0,6872) = 0,2150$

Node 1 = ~~0,2226 · 0,2~~ $= 0,04452 \times 0,2150 = 0,00957$

Node 2 = $0,02226 \times 0,2150 = 0,00479$

Node 3 = $0,11130 \times 0,2150 = 0,02393$

Node 4 = $0,04452 \times 0,2150 = 0,00957$

upstream gradient

$(0,00957 \times 0,3) + (0,00479 \times 0,5) + (0,02393 \times 0,3) + (0,00957 \times 0,3)$

$= 0,002871 + 0,001437 + 0.007179 + 0.002871 = 0,01436$

local gradient $= 0,2689 \cdot (1 - 0,2689) = 0,1966$

upstream x locals $0,01436 \times 0,1966 = 0,00282$

the yellow weight

upstream Gradient = 0,00282

local Gradient = 4

$\Rightarrow$ final Derivatives $= 0.00282 \times 4 = \boxed{0,01128}$

finnlensner